Exp1:

server:

```
#include<stdio.h>
#include<stdlib.h>
#include<sys/types.h>
#include<sys/socket.h>
#include<netinet/in.h>
#include<unistd.h>
#include<string.h>

int main()
{
        int server_sock;
        int n;
        server_sock=socket(AF_INET,SOCK_STREAM,0);
        struct sockaddr_in server_addr;
        server_addr.sin_family=AF_INET;
        server_addr.sin_port=htons(7249);
        server_addr.sin_addr.s_addr=INADDR_ANY;
        bind(server_sock,(struct sockaddr*)&server_addr,sizeof(server_addr));
        listen(server_sock,5);
        int client_sock;
        client_sock=accept(server_sock,NULL,NULL);
        int upper=1,lower=1,diagonal=1;
        recv(client_sock,&n,sizeof(n),0);
        int matrix[n][n];
        recv(client_sock,matrix,sizeof(matrix),0);
        for(int i=0;i<n;i++)
        {
         for(int j=0;j<n;j++)
         {
                if(i>j && matrix[i][j]!=0)
                {
                 upper=0;
                }
                if(i<j && matrix[i][j]!=0)
                {
                 lower=0;
                }
                if(i!=j && matrix[i][j]!=0)
                {
                 diagonal=0;
```

```
                }
            }
        }

        char result[50];
        if(diagonal)
          strcpy(result,"Diagonal Matrix");
        else if(upper)
          strcpy(result,"Upper Triangle Matrix");
        else if(lower)
          strcpy(result,"Lower Triangle Matrix");
        else
          strcpy(result,"None");
            send(client_sock,result,sizeof(result),0);
            close(client_sock);
            close(server_sock);
            return(0);
}


client:
#include<stdio.h>
#include<stdlib.h>
#include<sys/types.h>
#include<sys/socket.h>
#include<netinet/in.h>
#include<unistd.h>
#include<time.h>

int main()
{
            int sock,n;
            char result[30];
            sock=socket(AF_INET,SOCK_STREAM,0);
            struct sockaddr_in server_addr;
            server_addr.sin_family=AF_INET;
            server_addr.sin_port=htons(7249);
            server_addr.sin_addr.s_addr=INADDR_ANY;
            int conn_status=connect(sock,(struct sockaddr*)&server_addr,sizeof(server_addr));
            if(conn_status==-1)
                        printf("Error in making a connection\n");
            printf("Enter order of matrix:");
            scanf("%d",&n);
            int matrix[n][n];
```

```
        //srand(time(NULL));
        printf("Enter the Matrix:\n");
        for( int i=0;i<n;i++)
        {
        for(int j=0;j<n;j++)
        {
        scanf("%d",&matrix[i][j]);
        }
        }
        printf("Matrix:\n");
        for( int i=0;i<n;i++)
        {
        for(int j=0;j<n;j++)
        {
        printf("%d\t",matrix[i][j]);
        }
        printf("\n");
        }
        send(sock,&n,sizeof(n),0);
        send(sock,matrix,sizeof(matrix),0);
        recv(sock,result,sizeof(result),0);
        printf("server's answer: %s \n",result);
        close(sock);
        return(0);
}
```

Output:

```
//client
Enter order of matrix:3
Enter the Matrix:
5
4
5
0
8
3
0
0
4
Matrix:
5       4       5
0       8       3
```

0 0 4
server's answer: Upper Triangle Matrix

Exp 2:

server:

```
#include <stdio.h>
#include <string.h>
#include <stdlib.h>
#include <arpa/inet.h>
#include <unistd.h>

#define PORT 8080

void translate(char str[])
{
    struct
    {
        char shortf[20];
        char full[50];
    } dict[] =
    {
        {"tbh","to be honest"},
        {"sus","suspicious"},
        {"ig","I guess"},
        {"tbf","to be fair"},
        {"atm","at the moment"},
        {"irl","in real life"},
        {"lol","laughing out loud"},
        {"asap","as soon as possible"},
        {"omg","oh my God"},
        {"ttyl","talk to you later"},
        {"idk","I don't know"},
        {"nvm","never mind"},
        {"idc","I don't care"}
    };

    int size = sizeof(dict) / sizeof(dict[0]);

    char result[1024] = "";
    char temp[1024];
    strcpy(temp, str);
```